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About US

Appteknow is a premier training & placement institute. We are Pioneers in providing career-oriented courses to aspiring professionals from the year 2000. Our Institute is located in BTM layout, Bangalore since 24+ years. Our innovative training models yield 100% placement to our students.

Academic

Personality

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FULL STACK MERN
SYLLABUS



Developer Basics

- ·Explain client-server architecture.
- ·Use ipconfig (Windows) or ifconfig (Mac/Linux) to nd your local IP address.
- ·Access your local web server via the IP address.
- ·Learn what ports are and their role in IP communication.
- ·Use browser developer tools to inspect HTTP/HTTPS requests and responses.
- ·Understand DNS: How domain names map to IP addresses.



- ·Master terminal basics: ls, cd, mkdir, and le navigation.
- ·Study common project folder structures (e.g., src, public, config, models, routes).

Git & GitHub

- ·Install Git and congure username/email.
- ·Initialize a Git repository, stage changes, and commit code.
- ·Create a GitHub account and link your local repo to a remote.
- ·Use push, pull, and clone commands.
- ·Create branches, merge changes, and resolve conicts.
- ·Learn GitHub workows: Issues, Pull Requests, and code reviews.



HTML & CSS

- ·Write HTML documents with <html>, <head>, <body>, <meta>, and semantic tags.
- ·Use tags like <h1>, <p>, <a>, <img>, <ul>, and <form>.
- ·Apply CSS via inline, internal, and external stylesheets.
- ·Master selectors (tag, class, ID) and properties (margin, padding, flexbox).
- ·Build layouts with Flexbox and Grid.
- ·Add interactivity with :hover and :active.
- ·Implement responsive design with media queries.
- ·Inspect and debug CSS using browser tools.





JavaScript

- ·Include JavaScript in HTML via <script> tags.
- ·Learn syntax: let, const, loops, conditionals, arrays, and objects.
- ·Write functions (regular, arrow) and modular code.
- ·Manipulate the DOM with querySelector, innerHTML, and event listeners.
- ·Use ES6+ features: template literals, destructuring, and promises.
- ·Explore localStorage, cookies, and sessions.
- ·Parse and stringify JSON data.



Debugging Web Applications

- ·Debug JavaScript with Chrome DevTools.
- ·Inspect network requests and console errors.
- ·Use breakpoints and the “Sources” panel.
- ·Troubleshoot errors using StackOverow and GitHub.

REST APIs and Postman

- ·Understand REST principles and HTTP methods (GET, POST, PUT, DELETE).
- ·Test APIs with Postman: Send GET/POST requests, headers, and JSON payloads.
- ·Analyze status codes (200, 404, 500) and query parameters.

JWT & API Authentication

- ·Learn JWT structure (header, payload, signature).
- ·Generate and verify tokens in Node.js.
- ·Test authenticated endpoints in Postman with bearer tokens.

React.js

- Set up a React app with Vite or Create React App.
- Understand JSX syntax and component-based architecture.
- Create functional components and pass props.
- Manage state with useState and side effects with useEffect.
- Implement routing with react-router-dom.
- Fetch data from APIs using fetch or axios and display it dynamically.
- Build a project (e.g., to-do app, weather dashboard).



MySQL:

- ·Install MySQL and create tables with relationships (1:1, 1:many).
- ·Write SQL queries: SELECT, INSERT, JOIN, GROUP BY.
- ·Use indexes and optimize queries.
- ·Connect to Node.js with Sequelize ORM.

MongoDB:

- ·Install MongoDB and perform CRUD operations via the shell.
- ·Design document schemas with nested data.
- ·Use aggregation pipelines.
- ·Connect to Node.js with Mongoose ODM.



Node.js & Express

- ·Install Node.js and npm.
- ·Build a server with Express.js and handle routes (GET, POST).
- ·Use middleware like express.json(), cors, and morgan.
- ·Store environment variables with dotenv.
- ·Connect to MySQL (Sequelize) and MongoDB (Mongoose).
- ·Create REST APIs for CRUD operations.

Advanced Backend (Express)

- ·Structure projects using MVC architecture.
- ·Validate requests with express-validator or Joi.
- ·Handle le uploads with multer.
- ·Implement pagination, ltering, and sorting in APIs.
- ·Write unit tests with Jest or Mocha.
- ·Add error handling middleware and custom error classes.



Authentication & Security

- ·Hash passwords with bcrypt.
- ·Generate JWTs for user login/signup.
- ·Protect routes with role-based access (e.g., admin vs. user).
- ·Add OAuth 2.0 (Google/GitHub) using Passport.js.
- ·Store sessions in Redis or MongoDB.

Advanced React

- ·Manage global state with Redux Toolkit or Context API.
- ·Use advanced hooks: useReducer, useCallback, useMemo.
- ·Optimize performance with code splitting and lazy loading.
- ·Handle forms with Formik or React Hook Form.
- ·Add real-time features with WebSocket or Socket.io.



Deployment

- ·Containerize apps with Docker (Node.js + React).
- ·Deploy backend to Render or AWS EC2.
- ·Host React frontend on Vercel or Netlify.
- ·Set up CI/CD pipelines with GitHub Actions.
- ·Monitor apps with logging tools (e.g., Winston, Morgan).



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